



Research paper

Allometric relationships between cutaneous surface area and body mass in anuran amphibians



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ABSTRACT

Several allometric relationships exist in the literature describing the relationship between surface area (A_S) and body mass (M_B) for different taxa of Anura and these are frequently used in physiological studies. Here, we gathered literature data of A_S (cm^2) and M_B (g), added new original data, and allometric relationships between A_S and M_B were evaluated using linear and phylogenetic generalized least-square (PGLS) regressions. Data from 453 specimens from 44 species were included. Intraspecific allometric relations between A_S and M_B were determined for 18 species, with ten of those showing regressions not significantly different from the respective family regression; four species showing a significantly different y-intercept; and three species showing a significantly different slope. Only the families Bufonidae, Ranidae, and Hylidae were represented by several species (9, 11, and 12, respectively) and with larger specimens' numbers (54, 215, and 127, respectively). These three families exhibited significantly different OLS linear regressions on $\log_{10}$ -transformed data, with Hylidae showing the steepest (0.7735 ± 0.0110), Bufonidae an intermediate (0.6772 ± 0.0220), and Ranidae the lowest slope (0.6091 ± 0.0114). The relationship between A_S and M_B for Anura could be described by the linear regression $A_S = 9.8537 M_B^{0.6745}$ or by the PGLS regression $A_S = 8.7498 M_B^{0.685}$.

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1. Introduction

Body size varies considerably in anuran amphibians, the smallest known species are the Papua New Guinean *Paedophryne amauensis* Rittmeyer et al. 2012 [Microhylidae; adult snout-vent length (SVL) 7.0–8.0 mm; Rittmeyer et al., 2012] and the Brazilian flea-toads *Brachycephalus pulex* Napoli et al. 2011 and *B. sulfuratus* Condez et al. 2016 (Brachycephalidae; combined adult SVL 7.4–10.8 mm; Napoli et al., 2011; Condez et al., 2016), while the

largest species is the African *Conraua goliath* Boulenger, 1906 (Conrauidae), generally considered the largest frog in the world, with some individuals having a SVL of 32 cm SVL (Sabater-Pi, 1985). The integument of amphibians interacts with the environment, and is involved in gas, water, and ion exchange (Hillman et al., 2009). All of these processes depend on the surface area (A_S) available for passive diffusion and/or active transport.

Measurements of actual anuran A_S , however, are scarce and difficult to obtain due to the irregularly shape of these animal's bodies. To measure A_S in anurans, most studies removed the integument from euthanized animals, cut the skin into small pieces and determined the area of each piece (Rubner, 1883; Fry, 1913; Voit, 1930; Terroine and Delpech, 1931; Hutchison et al., 1968; McClanahan and Baldwin, 1969; Talbot and Feder, 1992; Dabés et al., 2012). Indirect methods to determine A_S have also been used,

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Table 1

Species investigated in the present study. Body mass (M_B) data given first as range followed by mean $\pm$ standard deviation. N gives the number of individuals sampled.

Species	N	M_B (g)	M_B (g)
<i>Dendropsophus branneri</i>	10	0.25–0.38	0.32 $\pm$ 0.04
<i>Dendropsophus decipiens</i>	10	0.22–0.46	0.27 $\pm$ 0.07
<i>Dendropsophus minutus</i>	10	0.66–1.07	0.78 $\pm$ 0.12
<i>Hypsiboas albomarginatus</i>	9	5.68–6.83	6.16 $\pm$ 0.42
<i>Leptodactylus natalensis</i>	7	6.03–9.22	7.28 $\pm$ 1.05
<i>Leptodactylus macrosternum</i>	10	20.88–55.43	37.31 $\pm$ 11.45
<i>Phyllodytes melanomystax</i>	10	0.90–1.17	1.03 $\pm$ 0.11
<i>Pristimantis paulodutrai</i>	10	0.58–1.34	0.84 $\pm$ 0.25
<i>Rhinella jimi</i>	10	123.03–522.94	338.40 $\pm$ 133.42
<i>Scinax auratus</i>	10	0.71–0.94	0.80 $\pm$ 0.08
<i>Scinax eurydice</i>	8	6.72–8.81	7.48 $\pm$ 0.64
<i>Scinax</i> sp. (<i>S. ruber</i> clade)	10	0.87–2.59	2.09 $\pm$ 0.54

such as measuring the capacitance of a copper-coated wax model (Buttemer, 1990) or, more recently, applying magnetic resonance imaging to estimate A_S in the Palmate Newt using 3D reconstructions (Wardziak et al., 2014).

From these databases allometric relationships between A_S and body mass (M_B) have been derived and are frequently being used to estimate physiological traits such as evaporative water loss (e.g. Ruibal and Hillman, 1981; Withers et al., 1982; Wygoda, 1984; Tattersall et al., 2006; Young et al., 2006; Tracy et al., 2007). Most studies have relied on the relationship between A_S and M_B for Anura established by McClanahan and Baldwin (1969), which was confirmed by Buttemer (1990). However, McClanahan and Baldwin (1969) used only three species of Bufonidae and one species of Scaphiopodidae in their study, weighing between 11.4 and 90.5 g, while Buttemer (1990) used two species of Australian tree frogs (Hylidae) weighing between 10.0 and 62.0 g and Talbot and Feder (1992) studied the bullfrog *Lithobates catesbeianus* (Shaw, 1802) (Ranidae) weighing between 40.0 and 514.8 g to establish the allometric relationship between A_S and M_B . Another attempt to examine this relationship by Hutchison et al. (1968) used regression analysis based on a larger dataset using bufonids (3 species), hylids (4 species), ranids (5 species) and scaphiopodids (1 species), ranging in overall body mass between 1.1 and 94.8 g. Comparing the commonly used regression given by McClanahan and Baldwin (1969) ($A_S = 9.9 M_B^{0.56}$) with the regressions given by Hutchison et al. (1968), the latter dataset yielded a regression line with greater scaling exponent when analyzing Bufonidae ($A_S = 9.246 M_B^{0.645}$), but a greater y-intercept when analyzing Anura ($A_S = 13.521 M_B^{0.579}$).

Rubner (1883) measured surface area for the first time in a frog (species not given), reporting A_S to be 116 cm² for an animal with M_B of 40 g. Applying the regression given by Hutchison et al. (1968) for Anura on a M_B of 40 g would result in an A_S of 114.44 cm², whereas McClanahan and Baldwin's regression would yield an A_S of 78.12 cm². Given these different allometric relationships between A_S and M_B reported in the literature and the resulting differences in estimates of anuran A_S , our aims in the present study were to compile data on anuran surface area and body mass, add new original data, and perform allometric regression analysis (ordinary and phylogenetic) on the largest datasets available to date, thereby establishing the best model at the level of species, family and order.

2. Material and methods

2.1. Data from this study

Surface area was determined in twelve species of anurans (Table 1). These animals were collected for a previous study (Dabés et al., 2012) under permits of the Instituto Brasileiro do Meio Ambi-

ente e dos Recursos Naturais Renováveis (License numbers 13001-1 and 2239548) and have been deposited in the amphibian collection of the Museu de Zoologia of the Universidade Federal da Bahia (UFBA). No animal has been killed for the sole purpose of determining its cutaneous surface area.

To determine A_S , animals were euthanized following the Guidelines for Use of Live Amphibians and Reptiles in Field and Laboratory Research (Beaupre et al., 2004) and preserved for up to one month in alcohol (70%). The integument (except hands and feet) was removed, stained in methylene blue, cut into pieces that would lay out on a flat surface, and all pieces were photographed together at 5-megapixel resolution using a digital camera at a constant distance. Total A_S of a specimen was determined from the digital photographs, transforming the pictures into black and white pixels and counting the number of black pixels representing integument using an *ad-hoc* developed program, Area (Camacho and Rocha, 2010). Anuran surface area was calibrated against a square of known area (1 cm²). Body mass of well-hydrated animals was determined during the experiments in Dabés et al. (2012) using a semi-analytical scale (Shimadzu). Bladder content was not removed before weighing.

2.2. Data from previous studies

To compare our results with previously published results, data of M_B and “ratio of surface of skin (in cm²) to body mass (in g)” of twenty species of anurans were taken from Table II of Czepek (1965) and used to calculate A_S (cm²). We also included in our analyses data of A_S and M_B taken from Fry (Table IV, 1913), from Voit (Table VI, 1930), from Terroine and Delpech (Tables I–IV, 1931), and from Chen et al. (2014), as given in their Supplemental S1. For the data from Voit (1930), we used his mean values of A_S , since he determined surface area both by skinning and by calculations using the animals' circumference. Furthermore, Fig. 18 from Hutchison et al. (1968), Fig. 2 from McClanahan and Baldwin (1969), Fig. 8 from Heatwole et al. (1969), and Fig. 2 from Talbot and Feder (1992) were extracted from pdf-files downloaded from the publisher's archives. Coordinates of data points were determined using the free software PlotDigitizer version 2.6.2. In all cases only data for whole animal surface area (except hand and feet) were included into our analysis (see Supporting Information 1).

We verified the quality of data obtained from the literature once the extracted data of A_S and M_B were log₁₀ transformed and analyzed with Prism Graphpad 6.0, performing an ordinary least square (OLS) linear regression analysis to compare the data extracted from the graphs published in Hutchison et al. (1968); Heatwole et al. (1969), McClanahan and Baldwin, 1969, and Talbot and Feder (1992) to the regression equations given in these papers.

2.3. Comparison and analyses of datasets

Data from Fry (1913), Voit (1930), Terroine and Delpech (1931), Czepek (1965) Hutchison et al. (1968), Heatwole et al. (1969), McClanahan and Baldwin (1969), Talbot and Feder (1992) and Chen et al. (2014) were pooled together for each family (Bufonidae, Hylidae, Ranidae, Scaphiopodidae) and again into one complete data set for all Anura. The pooled data of the four families and Anura as a whole were then compared to our data, grouped into Bufonidae, Craugastoridae, Hylidae, Leptodactylidae, and Ranidae, as well as Anura. A combined total of 433 data points from our study and from literature data, comprising 44 different species from 11 families, were used to analyze the relationship between A_S and M_B of Anura.

Data for individual species were analyzed separately when data from at least seven individuals were available to provide intraspecific allometric relations between A_S and M_B and to permit

comparisons with the regressions obtained of their respective families. Linear OLS regression analyses were carried out to compare the different data sets.

2.4. Comparative phylogenetic analysis

Phylogenetic information used to build the working phylogeny was based on the molecular study of Pyron and Wiens (2011) that was provided as supplementary material (topology and branch lengths) in Pyron and Wiens (2013). Our sampled data of A_S and M_B included 36 species, representing eleven anuran families, including the 12 species from the present study and 24 species from the literature from which we were able to identify the species. Therefore, we obtained a working phylogeny completely resolved (i.e. dichotomic) to the species level. For species with data for more than one individual, relationships within species level were randomly resolved rendering no polytomy (see Supporting information 2). Species with sampled data for which phylogenetic information was not available in Pyron and Wiens (2011) were replaced by the species most closely related. We summarized in (see Supporting Information 2) all species considered for comparative phylogenetic analysis (based on current taxonomy), number of individuals per species analyzed, data source, and the data matrix considered for phylogenetic analysis.

Comparative phylogenetic analysis were conducted using the packages Ape (Paradis et al., 2004), Caper (Orme et al., 2012), Geiger (Harmon et al., 2008) and Picante (Kembel et al., 2010) of the R platform (R Development Core Team, 2010). Unresolved relationships within species (i.e. data available for more than two individuals) were arbitrarily resolved with the package Ape (function 'multi2di' that transforms all multichotomies into a series of dichotomies branches of length zero). For the species *Odorrana grahami* (Boulenger, 1917) and *Rana temporaria* Linnaeus, 1758, for which data were available from a larger number of individuals, data were randomly sorted to provide an equitative sample size considering the available data for the remaining species (see Table 6, Supporting Information 2).

"Phylogenetic signal" reflects the "tendency for related species to resemble each other more than they resemble species drawn at random from the tree" (Blomberg and Garland 2002: 905). Phylogenetic signal was estimated for surface area (A_S) and body mass (M_B) using the index Pagel's λ (Pagel, 1999; Freckleton et al., 2002). Briefly, when $\lambda = 0$, which is equivalent to a "star phylogeny", i.e. absent phylogenetic signal, and when $\lambda = 1$, the traits are consistent with Brownian motion evolution based on branch lengths represented by the variance-covariance matrix. Values between zero and one indicate less phylogenetic signal than expected under a pure Brownian motion model (see Freckleton et al., 2002). Since estimates of the maximum likelihood of λ is not sufficient to decide whether a significant phylogenetic signal exists, we also compared this likelihood to the likelihood of models when $\lambda = 1$ and $\lambda = 0$ using a likelihood ratio test (LRT), as pointed out by Nunn (2011).

We performed a phylogenetic generalized least squares (PGLS) analysis (Martins and Hansen, 1997) to assess the association between A_S and M_B incorporating the phylogenetic history of the specimens (Martins and Hansen, 1997). To conduct the phylogenetic regression, the error structure of the residuals was optimized by simultaneously fitting the regression model via least squares (e.g. Grafen, 1989) through Pagel's λ in order to recover the performance of the best model (Ordinary Least Squares – OLS or PGLS) as suggested by Revell (2010). As outliers can seriously affect the parameter estimates of the regression model, we assessed its effects removing outliers with studentized residuals $\geq \pm 3$ as suggested by Jones and Purvis (1997) and implemented in accordance with Nunn (2011).

3. Results and discussion

3.1. Analysis of extracted values

The extraction of values of anuran A_S and M_B from published data generally resulted in regression lines not significantly different from the formulas given in the respective publications (for the detailed regression data see Supporting Information 3). Only the data for Pelobatidae extracted from Hutchison et al. (1968) and the data from McClanahan and Baldwin (1969) resulted in significantly different regressions when compared to the formulas given in each paper. Talbot and Feder (1992) also extracted values from Hutchison et al. (1968) and using least-squares regression found a significantly different allometric relationship for Pelobatidae, with values similar to the ones obtained in the present study (Slope: 0.56 ± 0.05 ; $\log_{10}$ -intercept: 1.053). Tracy (1976) obtained a regression similar to ours for ranids after extracting data from Hutchison et al. (1968): $A_S = 12.79 M_B^{0.606}$.

The regression obtained using values extracted from McClanahan and Baldwin (1969) showed a significantly steeper slope (0.6598 ± 0.0645) than in the original article (0.56). Whether this difference was based on an erroneous extraction of data from the graph given in the original paper, a different method used to calculate the regression, or whether a combination of both of these factors or some other mistake has occurred, is not known at present. However, slope and intercept of the regression based on the data extracted from McClanahan and Baldwin (1969) are similar to slope and intercept obtained when using the largest data set available for regression analysis of Bufonidae (Table 2).

Insofar as the majority of regression lines based on values of A_S and M_B extracted from the literature were not significantly different from the values for slope and intercept given in the original papers, we are confident that the method of extraction of A_S and M_B values from the literature resulted in reliable data that could be used for further analysis.

3.2. Analysis of individual species

Species-specific regression analyses were performed on 18 species belonging to six anuran families (Table 3, Supporting Information 4). Among these, the hylids *Dendropsophus decipiens* (Lutz, 1925), *D. minutus* (Peters, 1872), *Hypsiboas albomarginatus* (Spix, 1824), *Phyllodytes melanomystax* Caramaschi et al., 1992, *Scinax auratus* (Wied-Neuwied, 1821), *S. eurydice* (Bokermann, 1968), and *Scinax* sp. (*S. ruber* clade sensu Faivovich et al., 2005); *Leptodactylus macrosternum* Miranda-Ribeiro, 1926 (Leptodactylidae); *Pristimantis paulodutrai* (Bokermann, 1975) (Craugastoridae); and the bufonid *Rhinella jimi* (Stevaux, 2002) did not show regressions significantly different from the respective family regressions given in Table 2. *Anaxyrus boreas* (Baird and Girard, 1852) (Bufonidae), *Dendropsophus branneri* (Cochran, 1948) (Hylidae), and the ranids *L. catesbeianus* and *O. grahami* did show a significantly different y-intercept, whereas *Leptodactylus natalensis* Lutz, 1930 (Leptodactylidae) and the ranids *Pelophylax* sp. and *R. temporaria* did show a significantly different slope compared to the regressions of their respective families. While the difference between the regressions of *L. natalensis* and Leptodactylidae can be explained by the small M_B range available for this species, the regressions of *Pelophylax* sp. and *R. temporaria* seem to be heavily influenced by the smallest individual sampled of each species. When removing the smallest value from each of the latter species' dataset, both regressions change (slope 0.5413 ± 0.0826 , $\log_{10}$ -intercept 1.198 ± 0.1443 , $R^2 = 0.9148$ for *Pelophylax* sp. and slope 0.6407 ± 0.0483 , $\log_{10}$ -intercept 1.093 ± 0.0661 , $R^2 = 0.9024$ for *R. temporaria*), no longer being significantly different from the regression for all ranids.

Table 2
Linear OLS regressions of relationships of $\log_{10}$ -surface area (A_S) and $\log_{10}$ -body mass (M_B) of different anuran families, based on the combination of values extracted from the literature and values obtained in this study. N gives the number of individuals used to calculate the regression.

Taxon	Slope	$\log_{10}$ -Intercept	Antilog $_{10}$ -Intercept	R ²	N	Source
Bufonidae	0.6772 ± 0.0220	0.9007 ± 0.0390	7.9560 ± 1.0939	0.9478	54	Czopek (1965); Hutchison et al. (1968); McClanahan and Baldwin (1969); this study
Craugastoridae	0.8936 ± 0.1918	0.9984 ± 0.0293	9.9632 ± 1.0699	0.7308	10	This study
Eleutherodactylidae	0.6831 ± 0.0315	1.0180 ± 0.0165	10.4232 ± 1.0387	0.9853	9	Heatwole et al. (1969)
Hylidae	0.7735 ± 0.0110	0.9610 ± 0.0070	9.1411 ± 1.0162	0.9755	127	Czopek (1965); Hutchison et al. (1968); this study
Leptodactylidae	0.6652 ± 0.0279	0.9388 ± 0.0367	8.6856 ± 1.0882	0.9743	17	This study
Others	0.7298 ± 0.0476	0.8881 ± 0.0557	7.7286 ± 1.1158	0.9833	6	Czopek (1965); Voit (1930)
Ranidae	0.6091 ± 0.0114	1.1220 ± 0.0167	13.1826 ± 1.0416	0.9302	215	Fry (1913); Voit (1930); Terroine and Delpach (1931); Czopek (1965); Hutchison et al. (1968); Talbot and Feder (1992); Chen et al. (2014)
Scaphiropodidae	0.6012 ± 0.0484	1.0050 ± 0.0665	10.1158 ± 1.1653	0.9224	15	Czopek (1965); Hutchison et al. (1968); McClanahan and Baldwin (1969)
Anura	0.6745 ± 0.0059	0.9949 ± 0.0076	9.8537 ± 1.0179	0.9665	453	Combined data

The residual plots of the species-specific regressions (see Supporting Information 4) were mostly unbalanced on the x-axis due to the small variation in M_B of most species. However, since the residuals did not show heteroscedasticity and non-linear tendencies, we concluded that, using $\log_{10}$ -transformed data, linear regressions with $R^2 > 0.8$ sufficiently well described intraspecific allometry of surface area in the species studied here.

Hutchison et al. (1968) reported intraspecific regressions for the species studied therein, with the slopes varying from 0.266 to 0.582 among the Bufonidae, from 0.273 to 0.823 among Hylidae, from 0.606 to 0.725 among Ranidae, and being 0.714 for *Scaphiopus couchii* Baird, 1854 (Scaphiropodidae). The regressions given for *L. catesbeianus* by Hutchison et al. (1968) and by Talbot and Feder (1992) are similar to the one obtained in the present study, whereas the regression given by Hutchison et al. (1968) for *A. boreas* is somewhat different (slope 0.266, log-intercept 1.713), but in the same range as in the present study.

It seems clear that intraspecific relationships between A_S and M_B may be very different from a regression of the whole family, however, differences in slope and intercept may be largely caused by a limited range in sampled M_B , as indicated by a poor goodness of fit for some regressions. Larger species-specific datasets are generally needed to identify significant deviations from the family pattern of slope or intercept, thereby identifying morphological variations among different species.

3.3. Comparison of higher taxa

The data obtained in the present study were summarized in Fig. 1, yielding an allometric relationship described by the formula $A_S = 8.8227 M_B^{0.6782}$ for the 12 species of Anura measured by us. The regressions based on the combination of data from the present study and data extracted from the literature are given in Table 2 and Fig. 2. Only for the families Hylidae and Bufonidae the datasets available were large enough to permit comparison of the regressions based on literature data with our own data. Comparing the regression line obtained from A_S and M_B values of Bufonidae extracted from previously published sources (slope 0.6499 ± 0.0414 , $\log_{10}$ -intercept 0.9393 ± 0.0644) with our own data (slope 0.6230 ± 0.0439 , $\log_{10}$ -intercept 1.049 ± 0.1097), we found no significant difference in slope or intercept between both linear regressions, thus allowing us to calculate a slope of 0.6772 ± 0.0220 and a $\log_{10}$ -intercept of 0.9007 ± 0.0390 for the largest data set available by combining our data with the ones from the literature (Table 2 and Fig. 2).

The linear regression obtained for Hylidae based on previously published data (slope 0.6917 ± 0.0331 , $\log_{10}$ -intercept

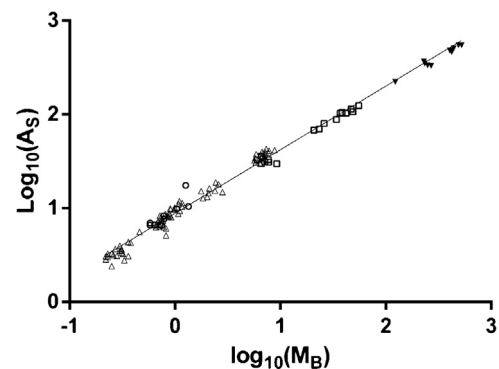


Fig. 1. $\log_{10}$ transformed values of surface area (A_S) plotted against $\log_{10}$ transformed values of body mass (M_B). Values are for Bufonidae ($\blacktriangledown$), Craugastoridae ($\circ$), Hylidae ($\triangle$), and Leptodactylidae ($\square$) obtained in the present study. The line represents the OLS linear regression ($A_S = 8.8227 M_B^{0.6782}$; $R^2 = 0.9896$; $N = 114$) for all anurans measured in the present study.

1.046 ± 0.0266) was not found to be significantly different from our regression line (slope 0.7512 ± 0.0123 , $\log_{10}$ -intercept 0.9465 ± 0.0061) regarding the slopes ($P = 0.0599$). The $\log_{10}$ intercepts were found to be significantly different ($P < 0.0001$). This difference may be explained by the fact that in the present study we used a large number of small specimens of Hylidae to determine A_S and M_B . Forty-five of our 77 specimens of hylids were weighing less than 1.0 g, whereas in previous papers, only one out of 50 hylids weighing less than 1.0 g had its A_S determined. Differences in size may explain the slightly steeper slope and the significantly different intercept in our data when compared to previous published data. Interestingly, the slopes for Hylidae obtained from the published data, from data obtained in our study, as well as from the combination of both data sets, were ≥ 0.7 and thereby greater than the slope (0.61) given by Hutchison et al. (1968) for Hylidae, as well as the slope of 0.66 given by Benedict (1932) for the relationship between A_S and M_B for vertebrates. Hylidae, therefore, seem to approximate roughly a $\frac{3}{4}$ power relationship between A_S and M_B (slope = 0.7735 ± 0.0110).

Regarding the families Craugastoridae, Eleutherodactylidae, Leptodactylidae, Ranidae and Scaphiropodidae, as well as the families Leiopelmatidae, Pipidae, Bombinatoridae and Pelobatidae, grouped here as 'Others', most slopes and intercepts were within the range of allometric relationships given in previous studies. Only the slope of Craugastoridae (0.8936) was much steeper than in other groups, but the goodness of fit ($R^2 = 0.7308$) was low for this group when compared to the other families. This difference seems

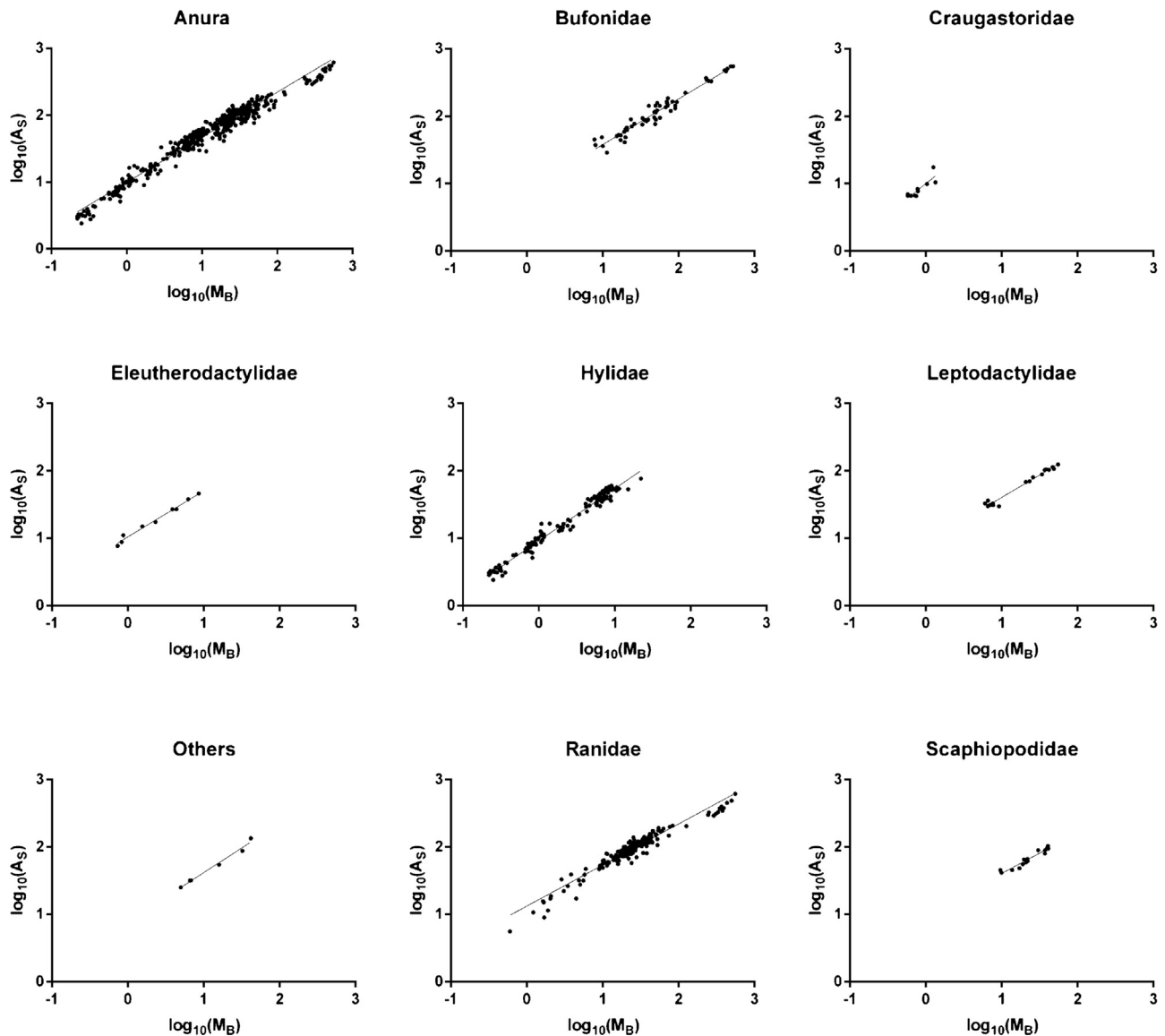


Fig. 2. $\log_{10}$ transformed values of surface area (A_S) plotted against $\log_{10}$ transformed values of body mass (M_B). Data for the different families represent the combination of values obtained from the literature with values sampled in the present study. Representatives of the families Leiopelmatidae, Pipidae, Bombinatoridae, and Pelobatidae were grouped together as 'Others' due to the low number of individuals sampled. The lines represent the OLS linear regressions for the different groups (see Table 2). Note that the data for Eleutherodactylidae and Craugastoridae were based on single species, *Eleutherodactylus portoricensis* and *Pristimantis paulodutrai*, respectively.

to have been caused by the very different A_S values found in the two largest specimens of Craugastoridae, all members of the species *P. paulodutrai*. One animal showed $M_B = 1.34$ g and $A_S = 10.44$ cm², while the other individual showed $M_B = 1.26$ g and $A_S = 17.51$ cm². If considering the latter value an outlier, the exclusion of this value resulted in the following relationship using a linear regression: $\log_{10}(A_S) = 0.6024 \log_{10}(M_B) + \log_{10}(0.9487)$. However, since there were no other data available regarding body mass and surface area for this species or family, we do not know for certain which one of these two individuals, if any, may be regarded as outlier. More data are necessary to obtain a better surface area to body mass relationship for this species and more species of the family are necessary to establish an allometric relationship for Craugastoridae.

We only analyzed Bufonidae, Hylidae, and Ranidae for significant differences because the other families were represented by low specimens' numbers and/or of a single species. The slopes of the three families were all significantly different from each other ($P < 0.0001$) with hylids showing the steepest, bufonids an intermediate, and ranids the lowest slope (Table 2, and Fig. 2). Because

the range in M_B available for the present analysis varied between the different families (Bufonidae 7.8–523.0 g; Hylidae 0.2–62.0 g; Ranidae 0.6–559.0 g), the relationships presented here could be altered with the inclusion of representatives of the smallest and largest members of each family. Taxonomists, however, normally do not include M_B when describing a new species and therefore data regarding the maximum M_B variation in the different anuran families are difficult to address. According to Vitt and Caldwell (2009) body size in bufonids may vary from small species such as *Dendrophryniscus carvalhoi* [Zecksohn, 1994, *Mertensophryne micranotis* (Loveridge, 1925), and *Pelophryne brevipes* (Peters, 1867), with combined adult SVL of <20 mm, to large species with a maximum SVL of up to 230 mm [*Rhinella marina* (Linnaeus, 1758)], whereas among the Hylidae adult SVL ranges from 12 mm [e.g., *Litoria microbelos* (Cogger, 1966), *Pseudacris ocularis* (Holbrook, 1838)] up to 135–140 mm adult SVL [e.g., *Litoria infrafrenata* (Günther, 1867), *Osteopilus vastus* (Cope, 1871)]. Ranids, on the other hand, are comprised of mostly medium to larger sized species (Vitt and Caldwell, 2009). These variations in SVL are probably sufficiently well

represented by the range in M_B sampled in the present study, suggesting that the addition of new data may not significantly change the allometric relationships described here for the three families.

An explanation for why the slopes for bufonids, hylids, and ranids were significantly different from each other could lie in the different body shapes seen in the three families. Bufonids are generally heavy-bodied, terrestrial animals with short legs (Wells, 2007), approximating with their barrel-shaped body the 2/3 allometric relationship of a regularly shaped object such as a sphere. Many hylids, on the other hand, do possess more arboreal habits, showing dorso-ventrally flattened bodies with long and slender legs, adaptations for adherence to the substrate and climbing, respectively (Tyler, 1976). The flattened body shape may well increase the A_S of a hylid when compared to a similar sized bufonid, with larger hylids possibly showing proportionally greater surface areas, resulting in a greater slope than bufonids. Species of the Ranidae mostly possess well-developed hind legs with much more skeletal muscle mass than in Bufonidae and Hylidae. These legs are used for forceful propulsion during swimming or jumping (Wells, 2007). Insofar as large ranids need more muscle mass to generate the force necessary for propulsion of their relatively heavier bodies than small ranids, their hind legs may contribute more surface area to total A_S , leading to a lesser slope than in bufonids. Interestingly, Hedrick et al. (2011) found significant differences between Bufonidae, Ranidae and Pipidae when studying the allometric relationships between M_B and lung volume and lung compliance, corroborating the existence of differences in morphological and physiological traits among different anuran families.

Comparing regressions from Anura of previously published data (slope 0.6125 ± 0.0088 , $\log_{10}$ -intercept 1.090 ± 0.0124) with our own data (Fig. 1; slope 0.6782 ± 0.00657 , $\log_{10}$ -intercept 0.9456 ± 0.0065), we found a significantly ($P < 0.0001$) different slope between both datasets. This difference, however, could also be explained by the large amount of small specimens sampled by us. In previous studies only five out of 319 anurans weighing less than 1.0 g have been studied with the smallest one weighing 0.5 g, whereas we included 52 out of a total of 114 specimens weighing less than 1.0 g, the smallest one with 0.22 g. Interestingly, the combination of all available data of anuran A_S and M_B yielded values for slope and intercept not significantly different ($A_S = 9.8537 \cdot M_B^{0.6745}$) from the values given by Benedict (1932; $A_S = 10 \cdot M_B^{0.66}$). The first physiologists studying the relationship between A_S and M_B used surface area data to correlate with heat loss and metabolic rate. Meeh (1879) established that the relationship between A_S and M_B could be described as $A_S = k M_B^{2/3}$, with k representing a constant that varies among different species with different body forms. Early researchers looked for variations in k , whereas variation in the exponent was usually not investigated. The k values obtained in earlier studies for anurans are, respectively: 9.92 (Rubner, 1883) [Rubner's published value is 4.62, however Voit (1930) used Rubner's original data of A_S and M_B to calculate a corrected k value], 11.2 (Fry, 1913), and 10.81 (Voit, 1930). These values are similar to the intercept obtained in this study by using the largest dataset available to date, indicating that the relationship of anurans as a group follows the general surface law of vertebrates. Since the slope obtained for Anura based on our data alone (Fig. 1) as well as the slope based on the combination of all data (Table 2) both follow the expected 2/3 relationship of surface area to volume, we are confident that the regressions obtained by us do adequately describe the relationship between A_S and M_B in Anura, despite using data sets of different origins.

Interestingly, the residual plots for both Anura and Ranidae (Fig. 3) indicate a non-linear pattern between A_S and M_B . Both very small and very large individuals may have their A_S overestimated using the allometric regressions given in Table 2 for Anura and Ranidae. Using a power function ($y = a + b x^c$) to fit a

curve through the original data resulted in the following relationships: Ranidae ($A_S = -12.184 + 23.107 M_B^{0.4854}$; $R^2 = 0.9522$) and Anura ($A_S = -4.335 + 15.790 M_B^{0.5544}$; $R^2 = 0.9591$; Fig. 4). The residual plots for the non-linear regression of Anura and Ranidae did not show heteroscedasticity, but were unbalanced on the x-axis and the values of the y-axis residuals were rather large when compared to the residuals from the transformed data.

When comparing the regressions obtained by combining all available data of the different families as well as for anurans, it seems clear that the size of the dataset is important for obtaining allometric relationships that best describe the relationship between A_S and M_B in anuran amphibians. Analyzing the relationship between basal metabolic rate and body mass in mammals, White and Seymour (2005) estimated that a subset of data from 150 species varying over 3–4 orders of magnitude is sufficient to describe the allometric relationship between both variables. We used 433 data points from 44 different species of anurans covering a body mass range between 0.22 and 559.0 g, a variation of more than 3 orders of magnitude. It may be criticized that we used multiple individual data from a single species to calculate a regression for a family and for Anura, instead of using a single mean value of A_S and M_B for each species. However, post-metamorphic anurans show differentiated allometric growth for different body parts, with juveniles possessing different body proportions than adults (Napoli and Caramaschi, 2000). Also, once reaching sexual maturity, amphibians may reduce growth rate, but do not cease growth (Guarino et al., 2003; Iturra-Cid et al., 2010; Arantes et al., 2015). Therefore, it is difficult to establish a representative mean value for a given species and the allometric regressions of A_S and M_B should take into account the juvenile forms, because the physiological processes linked to the integument are not restricted to adults and therefore we choose to calculate regressions for a family and for Anura using all individual data. Based on the White and Seymour (2005) suggestion, and since the vast majority of the roughly 6600 extant anuran species (Frost, 2016) may be included into the M_B range covered here, we conclude that the regressions obtained in the present study for Anura (Table 2) may describe sufficiently well the relationship between A_S and M_B , whereas specific families and or species may indeed show significant deviations from the anuran scaling relationship (Table 3).

3.4. Phylogenetic analysis

Phylogenetic signal was very high for both $\log_{10}$ - A_S and $\log_{10}$ - M_B , showing a strong phylogenetic association ($\lambda = 1.0$; Table 4). Thus, the phenotypic variance found in both traits does not differ of that expected under a Brownian motion model of evolutionary change (i.e. "a constant variance process" cf. Freckleton et al., 2002). Considering the phylogenetic relationships among the currently available dataset, i.e. 36 species assigned to 23 genera, and comprising eleven anuran families, Phylogenetic Generalized Least Squares (PGLS) analysis revealed that $\log_{10}(A_S)$ and $\log_{10}(M_B)$ were positively and highly correlated ($R^2 = 0.97$; $p < 0.0001$), with a $\log_{10}$ -intercept of 0.942 and a regression slope of 0.685 (see Table 5). Also, the relationship between $\log_{10}(A_S)$ and $\log_{10}(M_B)$ was highly associated with phylogeny ($\lambda = 0.998$; Table 5). PGLS analyses were also conducted separately for Bufonidae, Hylidae and Ranidae, providing regressions somewhat different (intercept or slope) than the regression for Anura (Table 5). Ranidae showed a regression slope similar to Anura (0.682 vs. 0.685 respectively), whereas Hylidae and Bufonidae (0.729 and 0.730, respectively) showed greater slopes than Anura. The $\log_{10}$ -intercepts were similar for Hylidae, Ranidae, and Anura (0.959, 0.999, and 0.942, respectively), but lower for Bufonidae (0.773).

Interestingly, the allometric regressions obtained using PGLS were very different from the OLS linear regressions obtained for

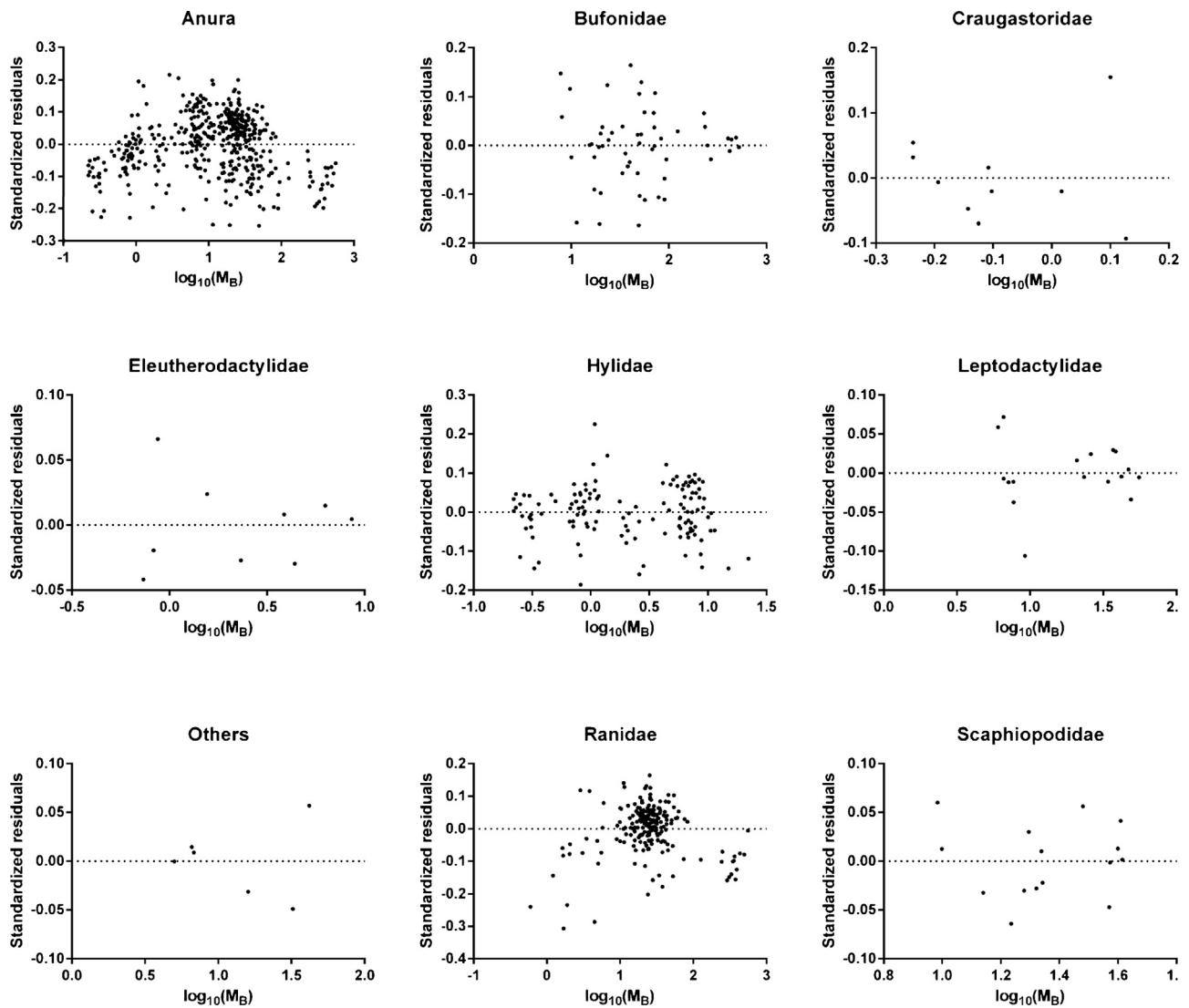


Fig. 3. Standardized residual plots of $\log_{10}$ transformed values of surface area (A_S) plotted against $\log_{10}$ transformed values of body mass (M_B). Data for the different families represent the combination of values obtained from the literature with values sampled in the present study. Representatives of the families Leiopelmatidae, Pipidae, Bombinatoridae, and Pelobatidae were grouped together as 'Others' due to the low number of individuals sampled. Note that the data for Eleutherodactylidae and Craugastoridae were based on single species, *Eleutherodactylus portoricensis* and *Pristimantis paulodutrai*, respectively.

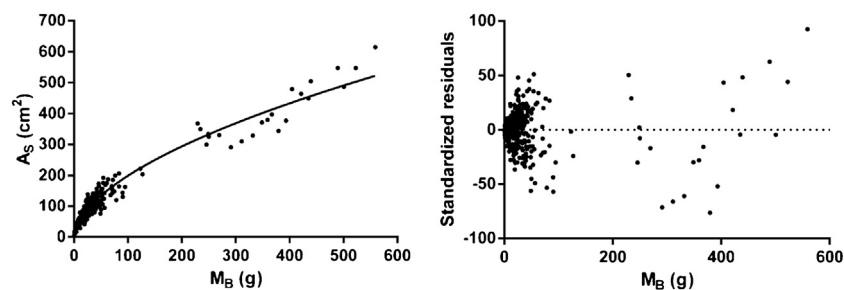


Fig. 4. Original values of surface area (A_S) plotted against body mass (M_B) for Anura (left). The curve describes a power function ($A_S = -4.335 + 15.790 M_B^{0.5544}$; $R^2 = 0.9591$). The plot on the right shows the standardized residuals of A_S plotted against M_B .

bufonids, hylids and ranids, but very similar to the linear regression obtained for Anura. Bufonids and ranids both showed lower intercepts and greater slopes, whereas hylids showed a similar intercept, but a lower slope, when comparing PGLS regression to OLS regression. These differences may be explained by the reduced dataset used to perform PGLS analysis and the small number of species (and consequently specimens) available to perform PGLS

on the Bufonidae, Hylidae and Ranidae families. However, since the PGLS as well as the linear regression for Anura were nearly identical despite using different datasets, we are confident that the OLS linear regression obtained using the largest dataset available for Anura best describes the relationship between A_S and M_B for this taxon.

Table 3
Linear OLS regressions of relationships of $\log_{10}$ -surface area (A_S) and $\log_{10}$ -body mass (M_B) of the individual species studied herein. N gives the number of individuals used to calculate the regression.

Species	Slope	$\log_{10}$ -Intercept	Antilog $_{10}$ -Intercept	R ²	N	Source
Bufonidae						
<i>Anaxyrus boreas</i>	0.4946 ± 0.0418	1.1390 ± 0.0683 ^b	13.7721 ± 1.1703	0.9656	7	McClanahan and Baldwin (1969)
<i>Rhinella jimi</i>	0.6230 ± 0.0439	1.049 ± 0.1097	11.1944 ± 1.2874	0.9619	10	This study
Craugastoridae						
<i>Pristimantis paulodutrai</i>	0.8936 ± 0.1918	0.9984 ± 0.0293	9.9632 ± 1.0699	0.7308	10	This study
Eleutherodactylidae						
<i>Eleutherodactylus portoricensis</i> ^c	0.6831 ± 0.0315	1.0180 ± 0.0165	10.4232 ± 1.0387	0.9853	9	Heatwole et al. (1969)
Hylidae						
<i>Dendropsophus branneri</i>	1.046 ± 0.3921	1.044 ± 0.1944 ^b	11.0662 ± 1.5646	0.4708	10	This study
<i>D. decipiens</i>	0.8391 ± 0.1053	1.018 ± 0.0614	10.4232 ± 1.1519	0.8881	10	This study
<i>D. minutus</i>	0.3592 ± 0.3917	0.8822 ± 0.0488	7.6243 ± 1.1190	0.0951	10	This study
<i>Hypsiboas albomarginatus</i>	0.6754 ± 0.4621	1.008 ± 0.3646	10.1859 ± 1.3153	0.2338	9	This study
<i>Phyllodytes melanomystax</i>	0.6563 ± 0.2992	0.9961 ± 0.0132	9.9106 ± 1.0308	0.3755	10	This study
<i>Scinax auratus</i>	0.7290 ± 0.3085	0.9666 ± 0.0324	9.2598 ± 1.0774	0.4111	10	This study
<i>S. eurydice</i>	0.5397 ± 0.2309	1.122 ± 0.2016	13.2434 ± 1.5907	0.4767	8	This study
<i>Scinax</i> sp. (<i>S. ruber</i> clade)	0.6350 ± 0.1139	0.9646 ± 0.0378	9.2172 ± 1.0909	0.7953	10	This study
Leptodactylidae						
<i>Leptodactylus natalensis</i>	−0.2294 ± 0.1827 ^a	1.701 ± 0.1571	50.2343 ± 1.4358	0.2397	7	This study
<i>L. macrosternum</i>	0.6104 ± 0.0463	1.028 ± 0.0721	10.6660 ± 1.1806	0.9560	10	This study
Ranidae						
<i>Lithobates catesbeianus</i>	0.7051 ± 0.0374	0.7743 ± 0.0937 ^b	5.9470 ± 1.2409	0.9727	12	Czopek (1965); Talbot and Feder (1992)
<i>Odorrana grahami</i>	0.5874 ± 0.0293	1.179 ± 0.0418 ^b	15.1008 ± 1.1009	0.8328	83	Chen et al. (2014)
<i>Pelophylax</i> sp.	0.7300 ± 0.0592 ^a	0.8608 ± 0.0959	7.2577 ± 1.2471	0.9682	7	Voit (1930); Czopek (1965)
<i>Rana temporaria</i>	0.7494 ± 0.0389 ^a	0.9416 ± 0.0520	8.7418 ± 1.1271	0.9490	22	Voit (1930); Czopek (1965)

^a indicates an intraspecific slope that is significantly ($p < 0.05$) different from the family regression given in Table 2.

^b indicates an intraspecific intercept that is significantly ($p < 0.05$) different from the family regression given in Table 2.

^c *Eleutherodactylus portoricensis* was not compared to the regression of the family, since the dataset was identical.

Table 4
Phylogenetic signal estimates (Pagel's λ) for the traits of $\log_{10}(A_S)$ and $\log_{10}(M_B)$ for 36 species, comprising eleven anuran family.

Trait	λ	($\lambda > 0$) p	($\lambda < 1$) p	Number of tips(individuals)
$\log_{10}(A_S)$ (cm ²)	1.00	<2.22e ^{−16}	1.00	207
$\log_{10}(M_B)$ (grams)	1.00	<2.22e ^{−16}	1.00	207

The p value reflects statistical significance based on the randomization test for phylogenetic signal.

$\lambda > 0$ ($p < 0.05$) indicates significant phylogenetic signal (PS) under a Brownian motion model of evolution.

$0 < \lambda < 1$ ($p < 0.05$) indicates less PS than expected under a pure Brownian motion model of evolution.

$\lambda = 1$ ($p > 0.05$) indicates the PS expected under a Brownian motion model of evolution.

Table 5
Correlated evolutionary change analysis between $\log_{10}(A_S)$ (Y) and $\log_{10}(M_B)$ (X) through Phylogenetic Generalized Least Squares (PGLS) for eleven anuran families, and for the families Bufonidae, Hylidae and Ranidae.

	Equation	Adjusted R ²	p	F-Statistic	DF	λ (p value)
Anura	$Y = 0.942 + 0.685X$	0.947	<2.2e ^{−16}	4829	167	0.998 ^a $\lambda = 0$ (2.22e ^{−16}) $\lambda = 1$ (0.932)
Bufonidae	$Y = 0.773 + 0.730X$	0.980	<2.2e ^{−16}	1276	25	0 $\lambda = 0$ (1) $\lambda = 1$ (0.022)
Hylidae	$Y = 0.959 + 0.729X$	0.964	<2.2e ^{−16}	2084	78	0.508 $\lambda = 0$ (1) $\lambda = 1$ (0.003)
Ranidae	$Y = 0.999 + 0.682X$	0.973	<2.2e ^{−16}	1896	52	1 ^a $\lambda = 0$ (3.40e ^{−0.9}) $\lambda = 1$ (1)

^a Correlation analysis (PGLS) showing significant phylogenetic dependency between traits.

3.5. Comparison with partial surface area data

Several authors have determined partial A_S data for anurans (Heatwole et al., 1969; Tracy, 1976; Buttemer, 1990; Prates and Navas 2009). Generally, the surface area exposed to air has been measured in animals using a water conserving posture (Pough et al., 1983), where the ventral surface is firmly attached to the substrate and the legs are pressed against the body, thereby exposing only dorsal and lateral surface areas to evaporative water loss. Withers et al. (1982) determined that the surface area exposed to air represents 2/3 of the total surface area of *Hyperolius nasutus* Günther, 1865 (Hyperoliidae), calculated by using the formula given by McClanahan and Baldwin (1969). This 2/3 estimate has been used frequently in studies concerning water balance in anurans.

Heatwole et al. (1969) determined total and partial surface areas for *Eleutherodactylus portoricensis* Schmidt, 1927 and according to their data, when this species assumes a water conserving posture, 47.3 ± 4.1% (Mean ± standard deviation) of its total surface area is exposed to air (calculated by extracting data of A_S and M_B from the original paper of individuals in a water conserving posture and comparing these values to total surface area; see Supporting Information 5). Using different methods, Buttemer (1990) and Prates and Navas (2009) also determined the surface area exposed to air in animals assuming a water conserving posture. Buttemer (1990) studied two species of hylids, *Litoria caerulea* (White, 1790) and *Litoria chloris* (Boulenger, 1892), whereas Prates and Navas (2009) studied the bufonid *Rhinella granulosa* (Spix, 1824). When extracting the values of dorsal surface area given in these publications and comparing them to total A_S values calculated using the formulas of

OLS linear regression for hylids and bufonids (see Table 2), the following percentages for dorsal surface area can be estimated: *Litoria caerulea* $32.5 \pm 2.5\%$, *L. chloris* $35.0 \pm 2.6\%$, and *Rhinella granulosa* $41.6 \pm 5.1\%$.

As no measured data of total A_S are available for the species studied by Buttemer (1990) and Prates and Navas (2009), and once the percentages of dorsal A_S estimated here are much smaller than the 2/3 estimate proposed by Withers et al. (1982), we consider our estimates to have to be treated cautiously. The best indication that the percentage of dorsal surface area exposed to air in an animal assuming a water conserving posture might be much smaller than previously anticipated has been presented by Heatwole et al. (1969).

Interestingly, Tracy (1976) determined the area in contact with the substrate in *Lithobates pipiens* (Schreber, 1782) (formerly *Rana pipiens*) in a stereotyped crouched position where the thighs are in contact with wet substrate (Dole, 1965). Using the data of M_B and partial A_S from Tracy (1976), one can calculate the total A_S of the animals used in his study by applying the formula given for Ranidae in Table 2. Based on these calculated values of total A_S , the surface area in contact with the substrate of animals in a stereotyped sitting position can be estimated as $8.1 \pm 1.5\%$ of total A_S . The stereotyped sitting position seen in *L. pipiens*, however, is not identical to a water conserving posture as being adopted occasionally in this species (Dole, 1965). It would be interesting to measure the ventral surface area in this species when adopting a water conserving posture and verify the increase in ventral surface area attached to the substrate, as well as to determine the amount of integument in contact with each other, such as the leg area pressed against the trunk.

Given the apparent variation in the estimates of the relationship between dorsal A_S and total A_S in anuran amphibians, it is necessary to gather more data in a variety of species before a general rule can be postulated. Recently, Lavy et al. (2015) presented a non-invasive method to determine surface area in benthic organisms using 3D computer modeling, which may be applied to other animals such as anurans as well, preserving thereby the integrity of the animals studied.

3.6. Conclusion

Taking into account that physiologists have, for over a century, been studying the surface area and its related functions, it is surprising that only a small number and relatively few species of anurans have been investigated in terms of their total and/or partial surface area and body mass. Furthermore, for only three families of anurans (i.e. Bufonidae, Hylidae, Ranidae) a suitable and representative number of data are available, thereby allowing estimation of the total surface area using one of the formulas presented here. The remaining 50 families of anurans (*sensu* Pyron and Wiens, 2011) are not sufficiently well represented by the general surface area to body mass relationship given for Anura, as seen for Bufonidae, Hylidae and Ranidae, and other families may deviate from the general anuran allometric relationship. However, the regressions obtained here for Anura as a whole, seem to better describe the allometric relationship between A_S and M_B than previously published regressions. When estimating A_S using the OLS regressions from our study and comparing these values to A_S calculated using the regression given by McClanahan and Baldwin (1969), one will find that the latter study will largely underestimate A_S . Allowing for a random variation of $\pm 10\%$ in A_S estimates, the OLS regressions presented herein and the regression from McClanahan and Baldwin (1969) will give similar estimates in A_S for Anura weighing between 0.5 and 2.4 g, for Bufonidae weighing from 2.7 to 14.6 g and for Hylidae ranging in M_B from 0.9 to 2.2 g. At M_B values lower or greater than the ranges given here, estimates of A_S using McClanahan and Baldwin (1969) will over- or underestimate A_S by more than 10%,

respectively. Regarding Ranidae, A_S estimates from McClanahan and Baldwin (1969) yield smaller values for every M_B value when compared to our OLS regression. Consequently, physiological data standardized to A_S values obtained through M_B measurements may significantly affect the results, if applying an erroneous regression. Following from our study we would recommend that, when studying physiological phenomena related to surface area, researchers should apply the appropriate regression or even better to actually measure the total and/or partial surface areas of the studied animals, especially when using species where no previous data are available. The actual investigation of a species' surface area related to physiology, as well as our general understanding of allometric relationships among anurans, would greatly benefit from these types of data collections.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.jcz.2016.04.007>.

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